Ajay Devidas Wanekar

M.Tech Energy Science and Technology

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SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of research experience in Artificial Intelligence and Machine Learning, specializing in Computer Vision, OCR-based text and character recognition, Vision-Language Models, and multimodal text-image understanding. Experienced in image preprocessing, dataset annotation, model benchmarking, and performance evaluation across challenging imaging conditions, along with hybrid semantic and keyword retrieval for matching extracted text against reference data.

EDUCATION

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | *LFOVIA Lab, IIT Hyderabad* **Sep 2024 – Jul 2025**

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding and OCR-based text and character recognition, and benchmarked task-specific models including RetinaFace, DeepFace, and Siamese architectures for targeted detection, face analysis, and visual tracking/re-identification.

Junior Research Assistant | *TIHAN, IIT Hyderabad:* **Nov 2023 – Jul 2024**

Worked on multimodal maritime perception by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | *WINET Lab, IIT Hyderabad:* **Jul 2023 – Nov 2023**

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJECTS

Multimodal Vision-Language Model Evaluation for Computer Vision and OCR Tasks:

- Evaluated Vision-Language Models (VLMs) for multimodal (text-image) understanding across computer vision tasks, including OCR-based text and character recognition, object detection, segmentation, age and gender classification, crowd analysis, and action/activity recognition, as alternatives to conventional task-specific models.
- Curated and annotated RGB and depth-mapping datasets, building structured preprocessing, labeling, and quality-validation pipelines that standardized inputs and improved model training consistency and reliability.
- Benchmarked and fine-tuned models using Python-based pipelines, evaluating detection and recognition performance (precision/recall, IoU) and optimizing data-processing workflows for improved scalability and computational efficiency.

Multi-Channel Conversational AI Platform:

- Built a multi-tenant conversational AI platform unifying voice, chat, and email with real-time STT/TTS pipelines and multilingual support.
- Developed a hybrid retrieval pipeline (semantic + keyword) that matches noisy, inconsistently phrased text queries against a reference knowledge base, with LLM-based intent classification, improving answer relevance by 18%.

M.TECH THESIS

Hybrid Machine Learning Modeling for Multi-Objective Optimization of a Non-Oxidative Methane Coupling DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict reactor performance from operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV).
- Achieved cross-validated R^2 values up to 0.996 using Gaussian Process Regression across six reactor-performance outputs, and identified Bayesian Optimization for experiment-guided process optimization.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL, PyTorch, TensorFlow, OpenCV, CUDA, NumPy, Pandas, Scikit-learn

Computer Vision & OCR: OCR (Text and Character Recognition), Image Preprocessing, Object Detection, Data Augmentation, Dataset Annotation, CNN, YOLO, DeepFace, RetinaFace

Multimodal AI & LLMs: Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), Transformer Architectures, Large Language Models (LLMs), LLM Fine-Tuning, RAG, Hybrid Semantic + Keyword Retrieval

Model Evaluation: Precision, Recall, IoU, Model Benchmarking

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs).” IEEE i-COSTE 2023, pp. 1−5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging.” IEEE i-COSTE 2023, pp. 1−6.

ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad - 2nd Place Winner

Languages: English, Hindi, Marathi, Japanese (Basic)